# Maternal infection during pregnancy and likelihood of autism and intellectual disability in children in Sweden: a negative control and sibling comparison cohort study

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*Lancet Psychiatry* 2022; 9: 782–91

Published Online September 7, 2022 http://.doi.org/10.1016/S2215-0366(22)00264-4

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For the Swedish translation of the abstract see Online for appendix 1

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## Summary

**Background** Maternal infections during pregnancy are associated with intellectual disability and autism in exposed children. Whether these associations are causal, and therefore should be targets of preventive strategies, remains unknown. We aimed to investigate these associations, to determine whether there is a causal role of maternal infection during pregnancy for children's risk of autism and intellectual disability, by accounting for unmeasured familial factors.

**Methods** We used a register-based cohort study design, and included children living in Stockholm County, Sweden, who were born in 1987–2010. We excluded children not born in Sweden, adopted children, and children with unknown biological mothers or fathers. Maternal infections during pregnancy, defined by ICD-8, ICD-9, and ICD-10 codes, were identified in the National Patient Register and Medical Birth Register. Children were followed up from birth to an outcome or a censoring event (death, migration from Stockholm, age 18 years, or Dec 31, 2016, whichever occurred first). The primary outcomes were diagnosis of autism or diagnosis of intellectual disability. We did a survival analysis to examine the association between inpatient and outpatient specialised care for any infection during pregnancy and likelihood of autism or intellectual disability in the child. To address potential residual confounding, we also estimated the relationship between maternal infection in the year preceding pregnancy as a negative control exposure and conducted a matched sibling analysis of sibling pairs who were discordant for autism or intellectual disability.

**Findings** 647 947 children living in Stockholm County were identified and, after excluding 97 980 children, we included 549 967 in the study (267 995 [48·7%] were female and 281 972 [51·3%] were male; mean age at censoring 13·5 years [SD 5·0; range <1 to 18]; 142 597 [25·9%] had a mother who was not born in Sweden). 445 (1·3%) of 34 013 children exposed to maternal infection during pregnancy were diagnosed with intellectual disability and 1123 (3·3%) with autism. 5087 (1·0%) of 515 954 unexposed children were diagnosed with intellectual disability and 13 035 (2·5%) with autism. Maternal infection during pregnancy was associated with autism (hazard ratio [HR] 1·16, 95% CI 1·09–1·23) and intellectual disability (1·37, 1·23–1·51) in exposed children compared with unexposed children. Maternal infection in the year before pregnancy (negative control exposure) was also associated with autism (HR 1·25, 95% CI 1·14–1·36), but was not associated with intellectual disability (1·09, 0·94–1·27). In sibling comparisons, the associations with maternal infection during pregnancy were attenuated for autism (HR 0·94, 95% CI 0·82–1·08; n=21 864), but not to the same extent for intellectual disability (1·15, 0·95–1·40; n=9275).

**Interpretation** Although infections in pregnant women are associated with both autism and intellectual disability in their children, the association with autism does not appear to reflect a causal relationship, but is more likely to be explained by factors shared between family members such as genetic variation or aspects of the shared environment. Thus, infection prevention is not expected to reduce autism incidence. For intellectual disability, unmeasured familial factors might not fully explain the observed associations, and a causal role of maternal infections cannot be excluded. Causal effects of specific but rare infections or infections not requiring health care contact cannot be excluded in either autism or intellectual disability.

**Funding** Swedish Research Council, Stanley Medical Research Institute, and Autism Speaks.

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## Introduction

Some infections (eg, *Toxoplasma gondii*, rubella, and cytomegalovirus) during pregnancy are human teratogens that can target the fetal brain causing malformations leading to behavioural, sensory, and cognitive deficits.[1] The gestational age of the fetus influences the outcome of such exposures, with earlier exposures associated with more severe outcomes.[2] Due to immunisation programmes against rubella, congenital infections are infrequent in most parts of the world, with only a small proportion of exposed children developing severe sequelae.[1] However, maternal infections during pregnancy by a broader range of common pathogens, not necessarily involving vertical transmission to the fetus, have been

## Research in context

### Evidence before this study

We searched PubMed from database inception to Jan 30, 2022, restricting the search to abstracts in English, using the terms ("Case-Control Studies"[Mesh] OR "Cohort Studies"[Mesh] OR "Prenatal Exposure Delayed Effects/epidemiology"[Mesh] OR "Registries"[Mesh] OR "Risk Factors"[Mesh]) AND ("Autism Spectrum Disorder"[Mesh] OR "Intellectual Disability"[Mesh] OR "Child Development Disorders, Pervasive"[Mesh]) AND ("Infections"[Mesh] OR "Pregnancy Complications, Infectious"[Mesh]) AND ("Pregnancy Complications"[Mesh] OR "Pregnancy"[Mesh]), to identify epidemiological studies of potential associations between maternal infections, prospectively recorded in health-care registers, and intellectual disability or autism. Increased chances of these outcomes, particularly autism, were reported in several, but not all, previous studies. A meta-analysis based on 36 studies reported a significantly increased risk (by 32%) of autism following maternal infection or fever during pregnancy. However, whether such associations are causal, and therefore should be targets of preventive strategies, remains unknown.

### Added value of this study

We found that maternal infections during pregnancy were associated with increased likelihood of autism and intellectual disability in exposed children. We also explored the potential for confounding of these associations by unmeasured familial factors, finding that maternal infections in the year before pregnancy (negative control exposures) were equally strongly associated with autism, but not with intellectual disability. When comparing full siblings, the association between maternal infections during pregnancy and intellectual disability was attenuated, although not to the same extent as for autism. Therefore, although our observations at the population level in Sweden are similar to those in previous studies, our negative control and sibling comparison analyses suggest that maternal infections requiring specialised care in pregnancy are not causally related to autism.

### Implications of all the available evidence

Although maternal infections requiring specialised care have been shown to be associated with autism in exposed children in previous studies and in our own results here, our results from the negative control and sibling comparison analyses suggest that maternal infections during pregnancy in general are not meaningful targets of preventive strategies. For intellectual disability, a causal role of maternal infections during pregnancy cannot be excluded. Moreover, this study does not contradict the well documented associations between specific but rare maternal infections during pregnancy, such as rubella, Zika, or cytomegalovirus, and children's likelihood of intellectual disability. It should be noted that our work does not exclude a causal role of other specific maternal infections during pregnancy and likelihood of autism or intellectual disability, and future studies will need to consider the potential effect of SARS-CoV-2, as our study took place before the global pandemic.

proposed to induce neurodevelopmental differences by other mechanisms, such as the transmission of a maternal immune signal across the placental–fetal interface. Such signals are hypothesised to change immune signalling pathways that are also involved in typical neurodevelopmental processes.[3] Experimental studies indicate that maternal infections or immune activation during pregnancy can induce autism-related behaviours and cognitive deficits in exposed animals.[4] Clinically, autism and intellectual disability show some diagnostic overlap,[5] and 25–30% of autistic people are also diagnosed with intellectual disability, despite positive phenotypic and genotypic associations between autism and high intelligence.[6]

Few epidemiological studies have investigated the potential relation between maternal infections with a broader range of common pathogens during pregnancy and intellectual disability in offspring.[7–9] The only studies to investigate the relationship between intellectual disability and common pregnancy infections considered urinary tract infections, and none accounted for familial confounding. In a register-based study including 376 579 children, Langridge and colleagues[7] reported a 36% increased odds of mild to moderate intellectual disability, but not severe intellectual disability, associated with maternal urinary tract infections during pregnancy, and associations of a similar magnitude were reported in two smaller cohort studies.[8,9]

Various epidemiological studies have reported associations between such maternal infections during pregnancy and risk of autism, although results have been inconsistent.[10–15] In a meta-analysis of 36 epidemiological studies, which included more than 6 million people, maternal infection or fever during pregnancy was associated with a 32% increase in odds of autism in the children.[15] In a previous cohort study of more than 2·3 million Swedish children, we reported a similar 30% increase in odds of autism associated with a maternal inpatient infection diagnosis during pregnancy, regardless of timing of infection or adjustments for various confounders.[11] A Swedish national cohort study of 1·3 million people reported an 81% increase in risk of autism associated with maternal inpatient diagnosis of severe infection and an 89% increase in risk of autism with maternal urinary tract infection, both of which were robust to moderate unknown confounding in quantitative bias analysis.[13] In a Danish study of 1·6 million children, which used registered maternal inpatient diagnoses of infections, there was no overall association with autism.[10] Similar to other studies,[10,12,14] significant associations

emerged, albeit inconsistently, after categorising the exposures according to timing, type of infection, or organ system affected.

See Online for appendix 2

Residual confounding by unobserved factors (eg, shared genetic liability between autism or intellectual disability and serious infections) has rarely been considered in previous studies. Host susceptibility to infections is associated with genetic factors,[16,17] and both autism and intellectual disability are associated with variants of genes related to the innate immune system.[18,19] A genetic correlation between treated infections and any psychiatric diagnosis has been reported.[18] Thus, there is a possibility that associations between maternal infections and autism or intellectual disability in observational studies might be partially explained by genetic confounding rather than representing causal relationships.[20]

We aimed to evaluate whether there is a causal role of maternal infections during pregnancy on children's likelihood of autism and intellectual disability, by accounting for unmeasured familial factors.

## Methods

### Study design and participants

We used a register-based cohort study design and included children living in Stockholm County, Sweden, who were born in 1987–2010. We excluded children not born in Sweden, adopted children, births not registered in the Medical Birth Register (MBR), children with unknown biological mothers or fathers, and children affected by a study outcome who were also diagnosed with chromosomal abnormalities or disorders of amino acid metabolism associated with intellectual disability or autism (appendix 2 p 9). To understand the potential for unmeasured confounding by familial factors, we additionally used two alternate comparison designs, using maternal infections during the year preceding pregnancy as a negative control, and comparing full siblings. Children with no siblings and sibling pairs without the relevant outcome were removed for the sibling analysis. Ethics approval was obtained by the Stockholm regional review board (DNR 2010/1185-31/5, DNR 2016/987-32). Individual informed consent was not required for this anonymised register-based study.

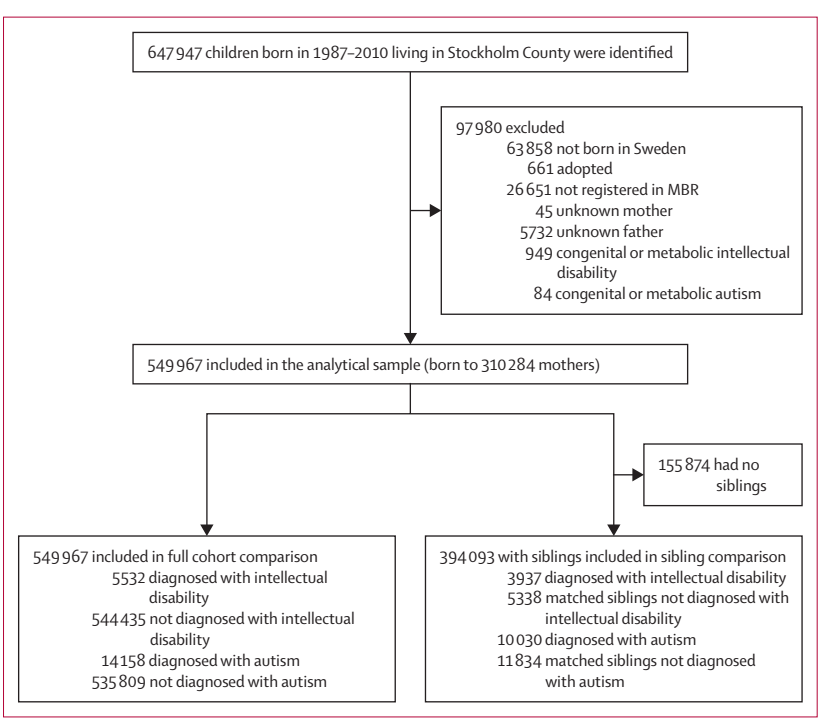


***Figure 1:*** **Study profile**
Congenital or metabolic intellectual disability or autism refers to children with intellectual disability or autism also diagnosed with congenital deformations, chromosomal abnormalities, or disorders of amino acid metabolism (appendix 2 p 9). MBR=Medical Birth Register.

### Procedures

All instances of inpatient or outpatient specialised care with a diagnosis of infection, defined by ICD-8, ICD-9 and ICD-10 codes (appendix 2 pp 10–11), were identified in the National Patient Register (NPR) and MBR, as detailed previously.[21] Infections were recorded if they were diagnosed during pregnancy, from the estimated conception date to the date of birth of the index child, or during the 1 year before the start of the index pregnancy for the negative control analysis. In pregnancies with missing conception dates (0·17%), the date of conception was imputed as 40 weeks before the index child's date of birth.

We considered other categorisations of infections for sensitivity analyses, including infections categorised as viral, bacterial, and other or unknown (appendix 2 pp 10–11),[11] and exposures categorised by trimester at diagnosis and restricted to those diagnosed in inpatient care, for comparability with previous studies.

Covariates were selected a priori on the basis of previously documented associations with intellectual disability or autism.[22,23] Children's covariates included sex, year of birth, birth order, and season of birth.[24] Parental covariates included parental region of origin (unknown or missing was coded as Swedish), quintiles of family income recorded at birth of the index child (calculated relative to the entire population of Sweden for the specific year to adjust for inflation, and adjusted by size of family; missing values were coded as the third quintile), highest education of the mother or father (missing was categorised as ≤9 years on the assumption that missing education information is translated as not finishing basic education or being a migrant without a registered education level), maternal BMI at first antenatal visit (in kg/m²; 27·2% missing information, coded as a separate category), parental age at birth, paternal infections, and lifetime history of maternal and paternal psychiatric history (appendix 2 p 9).

Children were followed up from birth to an outcome or a censoring event (death, emigration from Stockholm, age 18 years, or Dec 31, 2016, whichever occurred first).

## Outcomes

Diagnostic outcomes as of Dec 31, 2016 (the date of the latest register update) were defined by validated procedures covering all inpatient and outpatient pathways to care and diagnosis in Stockholm County using national and regional register resources (appendix 2 pp 10–11).[25] The primary outcomes were any diagnosis of autism or any diagnosis of intellectual disability.[25] Children who were diagnosed with both autism and intellectual disability would be included as cases in both analyses. Children were considered unaffected if they had not been diagnosed with the relevant outcome, although they could still be diagnosed with the other outcome. We also considered three mutually exclusive diagnostic outcomes: autism without co-occurring intellectual disability, autism with co-occurring intellectual disability, and intellectual disability without co-occurring autism. For these latter analyses, children were considered unaffected if they had not received any diagnosis of autism or intellectual disability. While we recognise that there is discussion among community members regarding terminology describing neurodevelopmental conditions,[26] here we chose to use the term unaffected to indicate children that did not receive the diagnosis of interest for each specific analysis, for clarity when comparing results.

## Statistical analysis

We calculated person-time from birth to diagnosis of an outcome or a censoring event. We used Cox regression models, assuming a proportional hazard from age 0 to 18 years (ie, the weighted average of the relative risk during childhood), to model potential associations between maternal infection during pregnancy and the outcomes, with the child's age as the underlying timescale. We used a robust sandwich estimator clustered on the mother to calculate SEs, corrected for the dependence between observations among children born to the same mother. We fitted models independently for the outcomes of any autism diagnosis, any intellectual disability diagnosis, autism without intellectual disability, autism with intellectual disability, and intellectual disability without autism, with the unaffected group for each analysis as already specified.

Minimally adjusted models (model 1; adjusted for the child's sex and birth year) were compared with models adjusted for the child's sex, birth year, birth order, and season of birth; maternal age and BMI; parental psychiatric history, income at birth, education level at birth, and regions of origin; and paternal infection during pregnancy (model 2 and model 3). Model 3 was additionally adjusted for maternal infections in the 1 year before pregnancy, to account for potential differences in susceptibility to infections or care-seeking behaviours between outcome groups.

To account for possible confounding by shared familial factors, we used two approaches (appendix 2 pp 1–2). First, as a negative control, we measured the risk of autism or intellectual disability associated with maternal infections in the 1 year before pregnancy (the analogous time period and seasonality

| | No maternal infection during pregnancy (n=515 954) | Maternal infection during pregnancy (n=34 013) | p value* |
|---|---|---|---|
| Autism | .. | .. | <0·0001 |
| No autism diagnosis | 502 919 (97·5%) | 32 890 (96·7%) | .. |
| Any autism diagnosis | 13 035 (2·5%) | 1123 (3·3%) | .. |
| Intellectual disability | .. | .. | <0·0001 |
| No intellectual disability diagnosis | 510 867 (99·0%) | 33 568 (98·7%) | .. |
| Any intellectual disability diagnosis | 5087 (1·0%) | 445 (1·3%) | .. |
| Mutually exclusive autism and intellectual disability diagnoses | .. | .. | <0·0001 |
| No autism or intellectual disability | 500 171 (96·9%) | 32 647 (96·0%) | .. |
| Autism with intellectual disability | 2339 (0·5%) | 202 (0·6%) | .. |
| Autism without intellectual disability | 10 696 (2·1%) | 921 (2·7%) | .. |
| Intellectual disability without autism | 2748 (0·5%) | 243 (0·7%) | .. |
| Sex | .. | .. | 0·082 |
| Female | 251 576 (48·8%) | 16 419 (48·3%) | .. |
| Male | 264 378 (51·2%) | 17 594 (51·7%) | .. |
| Single or multiple birth | .. | .. | <0·0001 |
| Singleton | 502 997 (97·5%) | 32 417 (95·3%) | .. |
| Twins | 12 652 (2·5%) | 1544 (4·5%) | .. |
| Triplets | 285 (0·1%) | 48 (0·1%) | .. |
| Quadruplets | 20 (<0·1%) | 4 (<0·1%) | .. |
| Birth order | .. | .. | <0·0001 |
| First born | 230 348 (44·6%) | 17 072 (50·2%) | .. |
| Second born | 188 740 (36·6%) | 10 749 (31·6%) | .. |
| Third born or later | 96 866 (18·8%) | 6192 (18·2%) | .. |
| Season of birth | .. | .. | 0·0002 |
| December–February | 121 061 (23·5%) | 7966 (23·4%) | .. |
| March–May | 139 537 (27·0%) | 9113 (26·8%) | .. |
| June–August | 134 147 (26·0%) | 8609 (25·3%) | .. |
| September–November | 121 209 (23·5%) | 8325 (24·5%) | .. |
| Maternal psychiatric diagnosis | .. | .. | <0·0001 |
| Not present | 295 148 (57·2%) | 15 942 (46·9%) | .. |
| Present | 220 806 (42·8%) | 18 071 (53·1%) | .. |
| Maternal pre-pregnancy infection† | .. | .. | <0·0001 |
| No pre-pregnancy infection | 501 688 (97·2%) | 30 908 (90·9%) | .. |
| Pre-pregnancy infection | 14 266 (2·8%) | 3105 (9·1%) | .. |
| Maternal age, years | .. | .. | <0·0001 |
| <25 | 75 181 (14·6%) | 6088 (17·9%) | .. |
| 25–29 | 152 878 (29·6%) | 9350 (27·5%) | .. |
| 30–34 | 180 442 (35·0%) | 11 125 (32·7%) | .. |
| 35–39 | 89 286 (17·3%) | 6039 (17·8%) | .. |
| >39 | 18 167 (3·5%) | 1411 (4·1%) | .. |
| Maternal BMI, km/m² | .. | .. | <0·0001 |
| Underweight (<18·5) | 13 964 (2·7%) | 943 (2·8%) | .. |
| Healthy weight (18·5 to <25) | 261 765 (50·7%) | 17 016 (50·0%) | .. |
| Overweight (25 to <30) | 73 385 (14·2%) | 6221 (18·3%) | .. |
| Obese (>30) | 24 292 (4·7%) | 2840 (8·3%) | .. |
| Missing‡ | 142 548 (27·6%) | 6993 (20·6%) | .. |

(Table 1 continues on next page)

| | No maternal infection during pregnancy (n=515 954) | Maternal infection during pregnancy (n=34 013) | p value* |
|---|---|---|---|
| (Continued from previous page) | | | |
| Maternal region of origin | .. | .. | <0·0001 |
| Sweden | 384 854 (74·6%) | 22 516 (66·2%) | .. |
| Nordics | 14 658 (2·8%) | 742 (2·2%) | .. |
| Western Europe | 22 941 (4·4%) | 1509 (4·4%) | .. |
| Africa | 22 860 (4·4%) | 2698 (7·9%) | .. |
| Asia and Oceania | 10 666 (2·1%) | 980 (2·9%) | .. |
| Middle East | 42 180 (8·2%) | 4115 (12·1%) | .. |
| Eastern Europe and Russia | 6896 (1·3%) | 556 (1·6%) | .. |
| South America | 10 825 (2·1%) | 890 (2·6%) | .. |
| Unknown or missing§ | 74 (<0·1%) | 7 (<0·1%) | .. |
| Family income quintile at birth | .. | .. | <0·0001 |
| Low (0–20%) | 72 189 (14·0%) | 5677 (16·7%) | .. |
| Medium-low (>20 to 40%) | 107 418 (20·8%) | 8337 (24·5%) | .. |
| Medium (>40 to 60%) | 112 120 (21·7%) | 7357 (21·6%) | .. |
| Medium-high (>60 to 80%) | 112 162 (21·7%) | 6591 (19·4%) | .. |
| High (>80 to 100%) | 111 413 (21·6%) | 6004 (17·7%) | .. |
| Missing¶ | 652 (0·1%) | 47 (0·1%) | .. |
| Highest parental education at birth, years | .. | .. | <0·0001 |
| <10 | 30 096 (5·8%) | 2794 (8·2%) | .. |
| 10–12 | 197 076 (38·2%) | 13 071 (38·4%) | .. |
| >12 | 287 618 (55·7%) | 18 056 (53·1%) | .. |
| Missing‖ | 1164 (0·2%) | 92 (0·3%) | .. |
| Paternal psychiatric diagnosis | .. | .. | <0·0001 |
| Not present | 374 296 (72·5%) | 23 308 (68·5%) | .. |
| Present | 141 658 (27·5%) | 10 705 (31·5%) | .. |
| Paternal infection during pregnancy | .. | .. | <0·0001 |
| No infection | 508 908 (98·6%) | 32 874 (96·7%) | .. |
| Pregnancy infection | 7046 (1·4%) | 1139 (3·3%) | .. |
| Paternal region of origin | .. | .. | <0·0001 |
| Sweden | 391 165 (75·8%) | 22 776 (67·0%) | .. |
| Nordics | 18 671 (3·6%) | 890 (2·6%) | .. |
| Western Europe | 15 018 (2·9%) | 1045 (3·1%) | .. |
| Africa | 18 916 (3·7%) | 2402 (7·1%) | .. |
| Asia and Oceania | 15 788 (3·1%) | 1515 (4·5%) | .. |
| Middle East | 35 645 (6·9%) | 3675 (10·8%) | .. |
| Eastern Europe and Russia | 10 992 (2·1%) | 810 (2·4%) | .. |
| South America | 9744 (1·9%) | 898 (2·6%) | .. |
| Unknown or missing§ | 15 (<0·1%) | 2 (<0·1%) | .. |

Data are n (%) or p values. *p values reported for $\chi^2$ test. †Pre-pregnancy infection was defined as infection during the analogous time period including the same seasons as the index pregnancy, but occurring in the year before the index child's conception. ‡Missing BMI observations were included in the model as its own category. §Missing observations were recorded as born in Sweden. ¶Missing observations were recoded as medium income. ‖Missing observations were recorded as <10 years of education.

***Table 1:*** **Characteristics of children in the study sample by exposure to maternal infection during pregnancy**

for the index pregnancy, but occurring in the year before the index child's conception). Second, we conducted a survival-based sibling analysis, in which we compared the occurrence of maternal infection among full-sibling pairs.

Sensitivity analyses investigated whether associations varied by trimester in which infection occurred, type of infection, or diagnoses restricted to inpatient care, and examined the potential for non-linear time-dependent associations (appendix 2 pp 1–2). We carried out a complete case analysis in which all children who were missing any covariate information were excluded from the analytical sample. To test for carryover effects in the sibling comparison,[27] sibling cohorts were divided into groups of older exposed and younger unexposed versus younger exposed and older unexposed.

All statistical analyses were performed using Stata/MP (version 15.1).

### Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

## Results

647 947 children living in Stockholm County were identified and, after excluding 97 980, the total analytical sample included 549 967 children (267 995 [48·7%] were female and 281 972 [51·3%] were male; mean age at censoring 13·5 years [SD 5·0; range <1 to 18]; figure 1). Information on ethnicity was not available, but 142 597 (25·9%) children had a mother who was not born in Sweden. The study data were analysed from May 28, 2021, to April 20, 2022. All covariates, except sex of the child, differed by exposure to maternal infections during pregnancy (table 1). Mothers who had infection during pregnancy were more likely to have a history of psychiatric illness and were more likely to have been treated for an infection in the year before pregnancy than mothers who did not have an infection during pregnancy (table 1). 445 (1·3%) of 34 013 children exposed to maternal infection during pregnancy were diagnosed with intellectual disability and 1123 (3·3%) were diagnosed with autism. 5087 (1·0%) of 515 954 unexposed children were diagnosed with intellectual disability and 13 035 (2·5%) were diagnosed with autism (table 1).

In the total analytical sample, all covariates differed by the presence of intellectual disability or autism (appendix 2 pp 12–15), including exposure to maternal infections during pregnancy (table 2). However, children diagnosed with either intellectual disability or autism were exposed to a maternal infection during pregnancy to a similar extent as their unaffected siblings (appendix 2 pp 16–17; table 2).

In the full cohort comparison (n=549 967), maternal infection during pregnancy was associated with both intellectual disability (hazard ratio [HR] 1·37, 95% CI 1·23–1·51) and autism (1·16, 1·09–1·23) in exposed children compared with unexposed children in the fully adjusted models (figure 2; appendix 2 p 3). We also observed associations between maternal infection during pregnancy and each of the mutually exclusive outcomes, with the highest likelihood observed for

intellectual disability without autism (HR 1·40, 95% CI 1·21–1·61), followed by autism with intellectual disability (1·35, 1·17–1·56), and autism without intellectual disability (1·13, 1·06–1·21; figure 3).

In the negative control exposure in the full cohort, maternal infection in the year before pregnancy was associated with autism (HR 1·25, 95% CI 1·14–1·36), but was not associated with intellectual disability (1·09, 0·94–1·27; figure 2). Maternal infection in the year before pregnancy was associated with autism without intellectual disability (HR 1·30, 95% CI 1·19–1·43), but was not associated with autism with intellectual disability (1·01, 0·81–1·27) or intellectual disability without autism (1·13, 0·92–1·39; figure 3).

After excluding children without siblings and sibling pairs without the relevant outcome, 394 093 children were included in the sibling comparison, including 9275 children in the comparison for intellectual disability (3937 with a diagnosis and 5338 siblings without), and 21 864 were included in the comparison for autism (10 030 with a diagnosis and 11 834 siblings without; figure 1). In the sibling comparison, the relative risk for intellectual disability was attenuated compared with the full cohort analysis and no longer showed a significant association (HR 1·15, 95% CI 0·95–1·40). The association between maternal infection during pregnancy and autism was completely attenuated in the sibling comparison (HR 0·94, 95% CI 0·82–1·08; figure 2; appendix 2 p 4). When implementing the non-stratified analytical models restricted to the sibling sample (n=394 093), our findings did not notably differ from the full population-based comparison (appendix 2 p 5).

For the mutually exclusive outcomes, the associations with maternal infection during pregnancy were attenuated for autism with intellectual disability (HR 1·20, 95% CI 0·92–1·58), intellectual disability without autism (1·18, 0·90–1·55), and autism without intellectual disability (0·89, 0·77–1·04) in the sibling analysis compared with the full cohort analysis (figure 3).

In the full cohort, there was no notable difference from the primary analyses when maternal infections were partitioned by type of infection (appendix 2 p 18) or trimester of pregnancy (appendix 2 p 19), or restricted to

| | Number of children | Children exposed to maternal infection |
|---|---|---|
| **Full cohort comparison (n=549 967)** | | |
| Intellectual disability | | |
| Diagnosis | 5532 | 445 (8·0%) |
| No diagnosis | 544 435 | 33 568 (6·2%) |
| Autism | | |
| Diagnosis | 14 158 | 1123 (7·9%) |
| No diagnosis | 535 809 | 32 890 (6·1%) |
| **Sibling comparison (n=394 093)** | | |
| Intellectual disability | | |
| Diagnosis | 3937 | 281 (7·1%) |
| Siblings without diagnosis | 5388 | 365 (6·8%) |
| Autism | | |
| Diagnosis | 10 030 | 693 (6·9%) |
| Siblings without diagnosis | 11 834 | 797 (6·7%) |

Data are n or n (%).

*Table 2:* **Exposure to maternal infection during pregnancy by diagnosis of intellectual disability or autism**

| | Cases | Person-years | HR (95% CI) Model 1 | Model 2 | Model 3 | Sibling model |
|---|---|---|---|---|---|---|
| **Any intellectual disability diagnosis** | | | | | | |
| No pre-pregnancy infection | 5352 | 7 216 766 | 1 (ref) | 1 (ref) | | |
| Pre-pregnancy infection | 180 | 190 116 | 1·32 (1·13–1·53) | 1·09 (0·94–1·27) | | |
| No pregnancy infection | 5087 | 7 022 404 | 1 (ref) | 1 (ref) | 1 (ref) | |
| Pregnancy infection | 445 | 384 478 | 1·63 (1·48–1·81) | 1·37 (1·24–1·52) | 1·37 (1·23–1·51) | |
| No pregnancy infection | 3656 | 106 572 | | | | 1 (ref) |
| Pregnancy infection | 281 | 6725 | | | | 1·15 (0·95–1·40) |
| **Any autism diagnosis** | | | | | | |
| No pre-pregnancy infection | 13 590 | 7 187 807 | 1 (ref) | 1 (ref) | | |
| Pre-pregnancy infection | 567 | 188 903 | 1·37 (1·26–1·49) | 1·25 (1·14–1·36) | | |
| No pregnancy infection | 13 034 | 6 994 459 | 1 (ref) | 1 (ref) | 1 (ref) | |
| Pregnancy infection | 1123 | 382 251 | 1·31 (1·23–1·40) | 1·17 (1·10–1·24) | 1·16 (1·09–1·23) | |
| No pregnancy infection | 9336 | 254 965 | | | | 1 (ref) |
| Pregnancy infection | 693 | 15 959 | | | | 0·94 (0·82–1·08) |

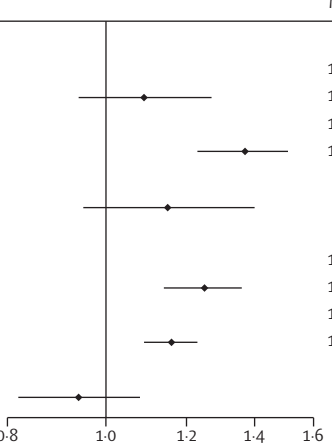


*Figure 2:* **The relationship between maternal infection during pregnancy and likelihood of intellectual disability and autism**
Person-years indicate the total person-time for the study sample according to exposure status. Pre-pregnancy infection was defined as infection during the analogous time period including the same seasons as the index pregnancy, but occurring in the year before the index child's conception. Model 1 was adjusted for child's sex and birth year. Model 2 was adjusted for child's sex, birth year, birth order, and season of birth; maternal age and BMI; parental psychiatric history, income at birth, level of education at birth, and regions of origin; and paternal infection during pregnancy. Model 3 was adjusted as model 2, but also adjusted for maternal pre-pregnancy infection. The sibling model was adjusted for child's sex, birth order, birth year, and season of birth; and paternal infection with model stratification on the nuclear family. The forest plot shows the point estimates for HR for each fully adjusted model on the log scale. Error bars are 95% CIs. HR=hazard ratio.

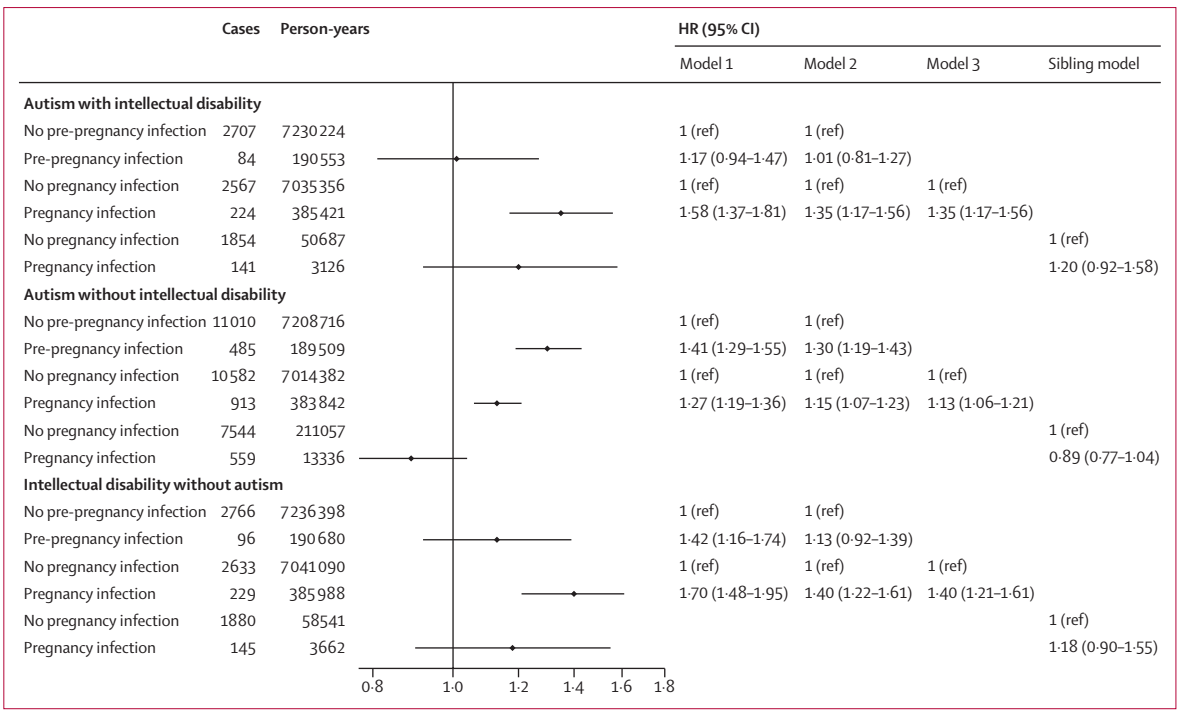


**Figure 3: The relationship between maternal infection during pregnancy and likelihood of the mutually exclusive outcomes of autism with intellectual disability, autism without intellectual disability, and intellectual disability without autism**

Person-years indicate the total person-time for the study sample according to exposure status. Pre-pregnancy infection was defined as infection during the analogous time period including the same seasons as the index pregnancy, but occurring in the year before the index child's conception. Model 1 was adjusted for child's sex and birth year. Model 2 was adjusted for child's sex, birth year, birth order, and season of birth; maternal age and BMI; parental psychiatric history, income at birth, level of education at birth, and regions of origin; and paternal infection during pregnancy. Model 3 was adjusted as model 2, but also adjusted for maternal pre-pregnancy infection. The sibling model was adjusted for child's sex, birth order, birth year, and season of birth; and paternal infection with model stratification on the nuclear family. The forest plot shows the point estimates for HR for each fully adjusted model on the log scale. Error bars are 95% CIs. HR=hazard ratio.

maternal infections diagnosed in inpatient care for both the full cohort (appendix 2 p 6) and the sibling comparisons (appendix 2 p 7). There were no apparent non-linear, time-dependent associations (appendix 2 p 3). We did not observe substantial differences between the results of the complete-case analysis and our primary analysis (appendix 2 p 20). We observed no indication of carryover effects in the sibling comparisons, with similar estimates when comparing older exposed or younger unexposed versus younger exposed or older unexposed sibling groups (appendix 2 p 21).

## Discussion

We investigated the associations between maternal infections during pregnancy and chance of autism and intellectual disability in their offspring in Sweden. In the general population, we observed an association between maternal infection during pregnancy and both autism and intellectual disability in exposed children, in concordance with previous register-based studies.[7,10,11,13] Moreover, we observed an association between maternal infection during the same period in the year before pregnancy and risk of autism, but this was not associated with intellectual disability. This discrepancy between autism and intellectual disability was highlighted in the analyses of mutually exclusive outcomes, where only autism without intellectual disability was associated with maternal infection in the year before pregnancy. Our analyses of full siblings suggest that familial factors—eg, genetic variation or shared environment—largely explain the association between maternal infection during pregnancy and autism observed in the general population. For intellectual disability, such factors might not fully explain the risk associated with maternal infection during pregnancy.

One previous study of autism compared exposure-discordant sibling pairs to controls for unmeasured familial confounders.[28] The authors found no association between gestational urinary tract infection and autism when comparing autistic children with general population controls or their non-autistic siblings. However, the study had a small number of exposed cases (n=17) and the follow-up period was short.

In our total study sample, we observed an association between a record of maternal infection during pregnancy and intellectual disability in exposed children. We observed no association between maternal infections during the year before pregnancy and intellectual disability, regardless of whether intellectual disability co-occurred with autism. The absence of an association with infections in the year

before pregnancy might indicate that the observed association in the full cohort comparison was not due to a general susceptibility to infections among mothers, but might rather be related to maternal infections during the pregnancy period specifically. This finding does not confirm, but is compatible with, a causal role of maternal infections in intellectual disability. In the sibling comparison, the associations between maternal infections and intellectual disability, regardless of co-occurring autism diagnoses, were attenuated and not significant. Overall, our results suggest that unmeasured factors shared between family members explain some, but potentially not all, of the observed association between maternal infections during pregnancy and risk of intellectual disability. Some specific maternal infections during pregnancy are known causes of intellectual disability, and sometimes also autism-related symptoms. In the current sample, such causal exposures were probably too rare to be detected.

Consistent with several epidemiological studies,[15] we observed an association between maternal infection during pregnancy and autism at the population level. However, an association of similar magnitude was also observed between maternal infections during the year before pregnancy and autism in their children, indicating that infection during pregnancy might in itself pose no specific change to the likelihood of autism. Additionally, associations were not observed in comparisons between sibling pairs, further supporting that these exposures are not causally related to autism. The absence of an association did not appear to be due the study being underpowered, as a non-stratified model (between-family comparison) restricted to the same smaller sibling population gave results similar to those observed in the total study population analysis. Taken together, these observations suggest that a general association between maternal infections during pregnancy and autism, particularly autism without intellectual disability, might be attributable to unmeasured confounding. Our analyses of mutually exclusive diagnostic subgroups indicate that confounding by familial factors might not fully explain the association with autism with intellectual disability. Family studies suggest that autism with intellectual disability is less familial than autism without intellectual disability.[29] The reported phenotypic and genotypic correlations between risk for maternal infections and autism might suggest that women with higher genetic liability for autism are more vulnerable to infections than the general population.[19]

A common limitation with exposure ascertainment of infections using medical records is that detection relies on contact with the health-care system. Previous register studies have relied solely on inpatient records. In this study, we combined inpatient and specialised outpatient care sources to capture a wider range of infections. Using this extended exposure ascertainment, we increased the proportion of infection exposures compared with previous register studies, which reported a prevalence of maternal infections in pregnancy of 1·3–3·4%.[10,11,13] Nevertheless, we could still only include mothers who sought specialised care, and we could not include infections treated in primary care. Potentially, the more inclusive ascertainment strategy might dilute relevant associations, if only the more severe infections were to have a harmful, causal effect on the developing fetus, and such infections were more likely to be treated in an inpatient setting. However, we observed similar results when comparing the broader exposure classification and inpatient-only diagnoses. We could not evaluate associations with specific or rare infections, infections not requiring contact with specialised health care, or infections that were not present in the population during the study period (eg, SARS-CoV-2). Due to health disparities across socioeconomic strata, mothers with low household income might be less likely to be ascertained as exposed if they were less able or less inclined to seek specialised care for infections. However, the universal Swedish health-care system provides care to all residents at a nominal cost, including a free programme of antenatal care, which would limit the impact of such ascertainment bias. Although we were not able to adjust for all confounding by sociodemographic factors, the sibling comparison allowed us intrinsically to account for factors that are shared within families, such as socioeconomic status or health-care seeking patterns. Missing observations in several covariates, including a relatively high proportion of missing data for maternal BMI, might introduce residual confounding even in our fully adjusted models. The degree of bias that could arise from our treatment of the missing observations is dependent on factors such as the proportion missing and the magnitude of the associations of the covariate with exposure and outcome (ie, potential for confounding). Because the proportion missing for all variables except maternal BMI was less than 0·5% and the magnitude of the associations with the outcomes was generally small, we assume that they were weak confounders.[22,23] The complete case analysis provided further reassurance that missing data were not likely to be a notable source of bias.

The use of register diagnoses to identify cases of intellectual disability and autism might miss less severe cases in the population. Such misclassification would drive estimated risks towards the null. However, a strength of our case ascertainment is that we use a well characterised cohort within a socialised health-care system with validated ascertainment procedures covering all pathways to habilitation services and care.[25] The prevalence of autism and intellectual disability in this study were in line with the international estimates (1·5% for autism and 1·0% for intellectual disability).[22,23]

There is evidence to suggest that the underlying cause of intellectual disability varies by grade of severity, with dominant non-inherited genetic variation explaining a large proportion of severe or profound cases, whereas the

milder cases seem to have a more complex multigenic background with potentially greater influence from environmental factors.[30] Treating intellectual disability as a single condition might preclude the detection of severity-specific causal effects, and severity grade would thus preferably be considered in the analyses. However, due to a large proportion of missing data (41·6%) on severity grade of intellectual disability and a low proportion (5·2%) of severe intellectual disability cases in this cohort,[31] our categorisations would be incomplete and our analysis would be underpowered.

Overall, the main strength of our study is the ability to compare estimates across different models—a general population estimate, a negative control exposure, and a sibling comparison—to assess and control for bias from unmeasured confounding factors. The negative control exposure design enables detection of unmeasured confounding by investigating an exposure with no plausible biological effect on the outcome of interest.[32] There is the possibility that long-term consequences of an infection in the year before pregnancy might affect the index pregnancy in some way. Mutual adjustments for differences in the rate of infections before or during pregnancy did not influence any of the associations, indicating that risks associated with these exposures were independent of each other. Analogous paternal exposures are often considered a negative control. In the case of paternal infections during the maternal pregnancy period, there is a possibility of disease transmission between the parents. Hence, paternal infections were only included as a covariate.

Previous studies have recognised the possibility of shared genetic liability between serious infections and neurodevelopmental conditions and have attempted to guard against such confounding by including diagnosed parental psychiatric history in their models.[10,11] Such measures are limited by their inability to capture subclinical traits and by uneven data quality over time and across populations (eg, for immigrants). By implementing a within-family approach, we were able to account for unmeasured familial factors.

Sibling comparisons are sensitive to bias from carryover effects,[27] measurement error, and unobserved confounders that are not perfectly shared by siblings.[33] There is a risk of measurement error for both exposure and outcome status, considering the inability to ascertain less severe cases, and time trends in the prevalence of autism, which could attenuate the within-pair estimates towards the null. However, changing the classification of infections (by restricting to inpatient diagnoses) yielded similar point estimates. We also found no evidence to suggest a presence of carryover effects of the exposure, in an analysis which also helps to reassure that the results are not strongly biased by time trends in autism diagnoses.

Although the results from the sibling comparison should be interpreted with some caution, they do align with the results of the negative control analyses. Together, these models indicate that the commonly observed association between maternal infection during pregnancy and exposed children's likelihood of autism might be confounded by unobserved factors shared within families, such as shared environment or genetic variation. The association between maternal infection during pregnancy and exposed children's likelihood of intellectual disability also appears to be explained by residual confounding, but potentially not to the same extent. It should be noted that our work does not exclude a causal role of specific maternal infections during pregnancy and likelihood of autism or intellectual disability, or both, and does not contradict the well-documented associations between, for example, maternal rubella or cytomegalovirus infections during pregnancy and children's risk for intellectual disability.

#### Contributors

MB drafted the first manuscript and contributed to study design and data analysis. HS drafted the first manuscript, and contributed to study design, data acquisition, and data analysis. HK contributed to study design, data analysis, and manuscript preparation. RMG contributed to study design, data acquisition, data analysis, and manuscript preparation. CD contributed to data acquisition, study design, and manuscript preparation. BKL contributed to study design and manuscript preparation. MB, HS, RMG, and CD had full access to and verified the data. All authors had final responsibility for the decision to submit for publication.

#### Declaration of interests

We declare no competing interests.

#### Data sharing

The Swedish health and population register data used in this study are available from Statistics Sweden and the Swedish National Board of Health and Welfare. The authors are not allowed to distribute the data according to the ethical approval for this study and the agreements with Statistics Sweden and the Swedish National Board of Health and Welfare.

#### Acknowledgments

This work was supported by grants from the Swedish Research Council (2016-01477, 2012-2264, and 523-2010-1052 [to CD]; 2017-02900 [to RG]), Autism Speaks (7618 [to BKL]), and the Stanley Medical Research Institute (to HK).

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